

Qualification Pack



Technician - Power System Transmission

QP Code: PSS/Q3403

Version: 1.0

NSQF Level: 4

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PSS/Q3403: Technician - Power System Transmission

Brief Job Description

The individual in the job role maintains sub-station, poles, towers and other related hardware in transmission system. He/she installs, maintains and repairs overhead and underground powerlines, cables and other associated equipment such as insulators, conductors, lightning arrestors, switches, metering systems, transformers and lighting systems.

Personal Attributes

The individual must be physically and mentally fit to be able to perform essential functions of the job safely. This will also include differently abled people who can perform the job with or without reasonable accommodations (modified practices.) The individual should be able to climb ladders, scaffolds, poles and towers of various heights. He/she should be able to work in confined spaces such as attics, manholes and crawlspaces. The individual should be able to read and understand written and verbal instructions respectively.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [PSS/N1331: Apply basic health and safety practices for power related work](#)
2. [PSS/N1336: Working effectively with others](#)
3. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)
4. [PSS/N3403: Perform repair & maintenance of power transmission lines and components](#)

Qualification Pack (QP) Parameters

Sector	Power
Sub-Sector	Transmission
Occupation	Operation and Maintenance - Transmission
Country	India
NSQF Level	4
Credits	14



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7413.0100
Minimum Educational Qualification & Experience	12th grade Pass (With Science Background) OR Completed 2nd year of the 3-year diploma after 10 OR Diploma (Pursuing 2nd year of 3-year regular diploma in relevant field after 10th) OR 10th grade pass (with two years of any combination of NTC/NAC/CITS or equivalent in relevant trade) OR 11th grade pass with 1 Year of experience relevant experience OR 10th grade pass with 2 Years of experience relevant experience OR Previous relevant Qualification of NSQF Level (3.5) with 1-2 Years of experience Relevant Experience OR Previous relevant Qualification of NSQF Level (3) with 3 Years of experience Relevant Experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	Electrical - 6 months
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	1.0
Reference code on NQR	QG-04-PW-00871-2023-V1-PSSC
NQR Version	1.0



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PSS/N1331: Apply basic health and safety practices for power related work

Description

This unit covers health, safety and security for power related work. This includes procedures and practices that candidates need to follow to help maintain a healthy, safe and secure work environment in a power plant, power station/substation or on the field while working on power equipment. It covers responsibilities towards self, others, organizational assets and the environment.

Scope

The scope covers the following :

- Follow workplace health and safety practices
- Follow fire safety practices
- Follow emergencies, rescue and first-aid procedures

Elements and Performance Criteria

Follow workplace health and safety practices

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)
- PC2.** identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)
- PC3.** use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment
- PC4.** follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)
- PC5.** ensure positive isolation of electrical equipment and system as per the given standards
- PC6.** identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.
- PC7.** use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)
- PC8.** use proper scaffolds, elevated platforms and ladders for working at height
- PC9.** inspect various tools and plants (T & P) routinely for any signs of oil, water leakages
- PC10.** store flammable materials and machine lubricating oil safely and correctly



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PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards

PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay

Follow fire safety practices

To be competent, the user/individual on the job must be able to:

PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)

PC14. use appropriate rescue techniques during fire hazard

PC15. use good housekeeping practices at all times to avoid accidental fire

Follow emergencies, rescue and first-aid procedures

To be competent, the user/individual on the job must be able to:

PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)

PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real

PC18. write an incident report and send the same to the authorized person

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. names (and job titles if applicable) and location of all documents along with the people responsible for health and safety in a workplace

KU2. possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity (such as drunkenness), health hazards (such as untreated injuries and contagious illness), not taking safety precautions

KU3. safe working practices when working with tools, machines

KU4. positive isolation of electrical equipment and system

KU5. use of emission and pollution control devices and measures taken to control pollution

KU6. preventative and remedial actions to be taken in the case of exposure to toxic materials solvents, flux, lead etc.

KU7. precautionary measures taken to prevent fire accident

KU8. various causes of fire (Causes of fires: heating of metal, spontaneous ignition, sparking, electrical heating, loose fires (smoking, welding, etc.), chemical fires, etc.)

KU9. different materials (sand, water, foam, CO₂, dry powder) used in fire extinguisher and proper method of using it to extinguish fire

KU10. various types of safety signs with their names

KU11. basic first aid treatment relevant to the condition e.g. shock, electrical shock, bleeding, breaks to bones, minor burns, resuscitation, poisoning, eye injuries basic techniques of bandaging

KU12. artificial respiration and the CPR Process



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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read & write an accident/ incident report in local language and/or English
- GS2.** read and comprehend basic content to read labels, charts, signages, manuals etc.
- GS3.** question co-workers appropriately and give clear instruction to them and sub ordinates in order to clarify instructions
- GS4.** remain congenial while discussing and debating issues with co-workers
- GS5.** observe appropriate protocols for communication based on situation, hierarchy, organizational culture and practice
- GS6.** plan and organize work schedule, work area, tools, equipment and materials to maintain decorum and for improved productivity
- GS7.** build customer relationships and use customer centric approach
- GS8.** seek assistance from other sources to resolve problem and report problems that cannot be resolved to the appropriate authority

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Follow workplace health and safety practices</i>	25	36	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions (Protective clothing: leather or asbestos gloves, flame proof aprons, flame proof overalls buttoned to neck, cuff less (without folds) trousers, reinforced footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, knee pads, particle masks, glasses/goggles/visors) (Equipment: hand and face shields, machine guards, residual current devices, shields, dust sheets, respirator)	3	5	-	-
PC2. identify job-site hazards and possible causes of risks/accident at the workplace (Hazards: electrical hazards, sharp edged and heavy tools, heated metals, oxyfuel and gas cylinders, welding radiation, hazardous surfaces, hazardous substances, physical hazards. Possible causes of risk and accident: physical actions, not following instructions, inattention, sickness and incapacity, health, not taking safety precautions)	3	2	-	-
PC3. use standard safe working practices when working at heights, confined areas, trenches and with electrical equipment	3	5	-	-
PC4. follow warning signs and symbols while working with electrical systems (Warning signs: Danger plate, high voltage area, out of service, etc.)	2	2	-	-
PC5. ensure positive isolation of electrical equipment and system as per the given standards	2	4	-	-
PC6. identify any abnormalities in electrical equipment or system viz temperature, pressure, flow of current etc.	2	4	-	-
PC7. use various methods of accident prevention in the work environment (Methods of accident prevention: use of equipment and working practices, safety notices, advice, instruction from colleagues and supervisors)	2	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC8. use proper scaffolds, elevated platforms and ladders for working at height	2	3	-	-
PC9. inspect various tools and plants (T & P) routinely for any signs of oil, water leakages	2	3	-	-
PC10. store flammable materials and machine lubricating oil safely and correctly	2	2	-	-
PC11. check that the emission and pollution control devices are working properly in line with environmental policy standards	1	3	-	-
PC12. inform the relevant authorities about any abnormal situation/behavior of any equipment/system without delay	1	3	-	-
<i>Follow fire safety practices</i>	6	12	-	-
PC13. use various appropriate fire extinguishers for different types of fires correctly (Types of fires: Class A, B, C, D and E)	3	5	-	-
PC14. use appropriate rescue techniques during fire hazard	2	5	-	-
PC15. use good housekeeping practices at all times to avoid accidental fire	1	2	-	-
<i>Follow emergencies, rescue and first-aid procedures</i>	8	13	-	-
PC16. administer appropriate first aid to the victims for the relevant injuries (Injuries: Bleeding, burns, choking, electric shock, poisoning etc.)	3	6	-	-
PC17. act promptly and appropriately to an accident situation and administer basic first aid including CPR to the victims of cardiac arrest due to electric shock or medical emergency in real	2	5	-	-
PC18. write an incident report and send the same to the authorized person	3	2	-	-
NOS Total	39	61	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1331
NOS Name	Apply basic health and safety practices for power related work
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1.5
Version	2.0
Next Review Date	31/08/2026



Qualification Pack

PSS/N1336: Working effectively with others

Description

This unit covers basic practices that improve effectiveness of working with others in an organizational set-up

Scope

The scope covers the following :

- Work effectively with others
- Respect diversity

Elements and Performance Criteria

Work effectively with others

To be competent, the user/individual on the job must be able to:

- PC1.** accept information and instructions from the supervisor and fellow workers and seek clarification when required
- PC2.** pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt
- PC3.** help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency
- PC4.** use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism
- PC5.** listen attentively and respond appropriately while interacting with others at work
- PC6.** act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)
- PC7.** escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict

Respect diversity

To be competent, the user/individual on the job must be able to:

- PC8.** respect gender diversity in the workplace
- PC9.** use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive
- PC10.** communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture
- PC11.** recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
- KU2.** reporting structure, inter-dependent functions, lines and procedures in the work area
- KU3.** relevant people and their responsibilities within the work area
- KU4.** escalation matrix and procedures for reporting work and employment related issues
- KU5.** key elements of active listening and assertive communication
- KU6.** importance of teamwork in organizational and individual success
- KU7.** ethics and disciplined behavior for professional success
- KU8.** expressing and addressing grievances appropriately and managing interpersonal conflict effectively
- KU9.** harassment and discrimination based on gender, disability, caste, religion and culture and how to recognize it.
- KU10.** organizational policy/ies for harassment and discrimination

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write clear and legible notes to self, colleagues and seniors to pass messages, keep records, prepare to-do lists, take down instructions
- GS2.** read basic terms and terminologies to accurately interpret work related documents, labels, supervisor instructions in the local language
- GS3.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the product, production plans and other associated requirements
- GS4.** display active listening skills while interacting with co-workers and other in the workplace
- GS5.** follow organization's rule-based decision-making process
- GS6.** use appropriate planning to maintain a smooth relationship with fellow team members
- GS7.** meet customer requirements
- GS8.** critically evaluate operation parameters in relation to system normality

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work effectively with others</i>	18	32	-	-
PC1. accept information and instructions from the supervisor and fellow workers and seek clarification when required	2	4	-	-
PC2. pass the information to the persons to all those concerns within agreed timeframe in a manner they can understand and confirm its receipt	1	4	-	-
PC3. help & assist fellow workers in performing their tasks in a positive manner so as to maximize efficiency	2	4	-	-
PC4. use appropriate tone, pitch and language while communicating to convey politeness, assertiveness, care and professionalism	4	6	-	-
PC5. listen attentively and respond appropriately while interacting with others at work	3	6	-	-
PC6. act responsibly and demonstrate disciplined behaviour at the workplace (Disciplined behaviour: Punctuality, completing tasks within agreed timescales, ensuring quality standards, not gossiping and wasting time, honesty, etc.)	3	5	-	-
PC7. escalate grievances and problems to the appropriate authority as per the organizational procedure to resolve them and avoid conflict	3	3	-	-
<i>Respect diversity</i>	16	24	-	-
PC8. respect gender diversity in the workplace	4	6	-	-
PC9. use respectful verbal, non-verbal and written communication that is gender, disability and culturally inclusive	4	6	-	-
PC10. communicate with everyone without any personal bias based on gender, disability, caste, religion, colour, sexual orientation and culture	4	6	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. recognize indicators of harassment and discrimination based on gender, disability, caste, religion, colour, sexual orientation and culture at workplace and follow organisational policy for reporting the same	4	6	-	-
NOS Total	34	56	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N1336
NOS Name	Working effectively with others
Sector	Power
Sub-Sector	Generic
Occupation	Generic
NSQF Level	2
Credits	1
Version	2.0
Last Reviewed Date	22/10/2024
Next Review Date	22/10/2027
NSQC Clearance Date	22/10/2024



Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values - Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:



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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e-mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.



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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings



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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026



Qualification Pack

PSS/N3403: Perform repair & maintenance of power transmission lines and components

Description

This unit is about the competencies required for repair and maintenance of sub-stations and power transmission lines.

Scope

The scope covers the following :

- Perform preparatory activities for repair and maintenance of power transmission lines.
- Identify tools and equipment required for repair and maintenance of power transmission lines.

Elements and Performance Criteria

Prepare for repair and maintenance of Power Transmission lines

To be competent, the user/individual on the job must be able to:

- PC1.** identify various types of circuits and their components correctly
- PC2.** identify and acquire correct tools, equipment and instruments required for various aspects of repair, maintenance, assessment and inspection of transmission lines and components
- PC3.** identify hazards in trimming trees such as limits of approach, public safety and step and touch potential access
- PC4.** conduct site inspection for emergency cases following established procedures
- PC5.** climb tower while following all necessary safety procedures and using ppe
- PC6.** identify various types of circuits accurately
- PC7.** ensure the tools and equipment are well maintained, calibrated and approved for use
- PC8.** use transmission line tools, equipment and hardware in line with job requirements for maintenance operations
- PC9.** prepare and maintain the work area as per procedure or operation specification
- PC10.** obtain work permit to proceed with work from the appropriate personnel in accordance with the organizational procedure
- PC11.** switch off, isolate, discharge and earth (side) line cables

Perform maintenance of power transmission lines as job specifications and requirements

To be competent, the user/individual on the job must be able to:

- PC12.** perform off-line overhead line maintenance procedure
- PC13.** perform off-line underground line maintenance procedure
- PC14.** ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required
- PC15.** install components on transmission lines including gang operated air brake switches for transmission lines, controlled breakers, ground switches, capacitor stations, insulator pressure washing, submarine and underground transmission cable, grid interconnections, etc.



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- PC16.** select and use test equipment such as tong testers, clip-on meter, multimeters, fault indicators meggers and voltmeters to verify fault and integrity
- PC17.** document switching procedures with all relevant details clearly and accurately
- PC18.** repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers
- PC19.** evaluate the work carried out by the team members and ensure it is as per the standard requirements and provide any feedback in a timely, polite and supportive manner
- PC20.** report any issues along with the remedial actions such as repairs or replacements, and estimated repair time to the relevant authority
- PC21.** replace pole as per standard procedure, where required
- PC22.** carry out guy and anchor replacement on various structure types: wood, steel, various lines voltages
- PC23.** carry out conductor patch and splice repair on single conductor, bundled conductor of various sizes and line voltages
- PC24.** replace components such as transformers, ct (current transformer), cvt (capacitive voltage transformer), la (lightening arrestor), breakers, towers, conductors, disconnects, timber or x-arm, conductors, poles, switches, elbows, terminations and insulators safely
- PC25.** replace other line components due to damage or unsuitability as per standard procedure, where required
- PC26.** replace underground cables, as per standard procedures where required

Perform post repair and maintenance activities

To be competent, the user/individual on the job must be able to:

- PC27.** restore the system to normal operating status by using switching procedures
- PC28.** identify problems that are not within one's own capability or limits of authority and seek support from authorized persons for resolving the same
- PC29.** leave the work area in a safe and tidy condition on completion of the repair and maintenance activities
- PC30.** refer unresolved job-related problems to the appropriate personnel for support
- PC31.** monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

1. relevant legislation, standards, policies, and procedures followed in the company relevant to own employment and performance conditions
2. relevant health and safety requirements applicable in the work place
3. own job role and responsibilities and sources for information pertaining to employment terms, entitlements, job role and responsibilities
4. reporting structure, inter-dependent functions, lines and procedures in the work area
5. how to engage with specialists for support in order to resolve incidents and service requests
6. importance of working in clean and safe environment practices and procedures
7. relevant people and their responsibilities within the work area
8. escalation matrix and procedures for reporting work and employment related issues



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9. principles of electricity
10. common electricity terminology and correct interpretation of the same
11. elements of the power system elements: generation, transmission, transmission metering, etc. elements: generation, transmission, transmission metering, etc.
12. different types of material and accessories used in power t&d materials and accessories: support (towers-steel, cement, wooden), materials and accessories: support (towers-steel, cement, wooden),
13. sizes and current carrying capacity of different types of conductors
14. conductor accessories: binding tape, binding wire, p.g. clamp, t clamp etc.; insulators - pin, disc, shackle, guy etc., lt switchgears, cross arms, stay sets, go switches etc. types of cross arms, etc.
15. tools and equipment: plier, screwdriver, wrench set, hammer, drilling machine, hacksaw/cutting tools, measuring tape, pulleys (force pulley with sling), tommy bar, crimping machine, round / flat file, earth rod i.e. discharge rod), leakage current monitoring kit
16. specific health and safety precautions which must be taken when carrying out transmission lines repair and maintenance work especially live line orequipment
17. various types of circuits and equipmenttypes: ct, pt, control, indication & annunciation circuits
18. line diagrams, maps and circuitry
19. troubleshooting and repair methods for various components of transmission lines
20. fault indicators for various components of the transmission lines
21. overhead transmission system apparatus such as regulators and reclosers
22. overhead transmission system standards
23. access points such as vaults, open trenches and manholes
24. underground transmission system apparatus such as transformers, switching cables and junction boxes
25. cable locating and fault detecting equipment
26. co-existing underground utilities
27. types and sizes of conductors and cables
28. causes of conductor damage
29. classification of conductor and insulator damage including fretting, abrasion,
30. fatigue breaks, tensile breaks
31. need for an authorized permit on 11 kv and above voltage line, if the line is under own authority, then take self-permit
32. hazards associated with carrying out power line maintenance and how they can be minimized
33. importance of ensuring that tools and equipment are suitable, well maintained, calibrated and operating effectively
34. importance of following good housekeeping and fire prevention procedures
35. importance of following job instructions and defined maintenance procedures
36. material preparation methods and techniques to be undertaken, prior to using for testing and maintenance activities
37. preparation of equipment for testing and repair activities
38. components of transmission lines



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39. procedures for handling transmission line components with imperfections/defects that cannot be removed/repared and how can they be minimized
40. problems and conditions which render electrical poles or towers in need of maintenance or replacement
41. importance of leaving the work area and equipment in a safe and clean condition on completion of the repair and maintenance activities
42. importance of reporting problems in a timely manner
43. methods and parameters to check quality of line components against required quality standards
methods: visual inspection, binoculars, measuring tape, use of instruments
44. calibration schedule of all equipment used in inspection, repair and maintenance activities
45. principles and practices of electrical safety
46. standard procedures how to deal with electric shocks and electrocutions to rescue and minimize damage and harm
47. safety concerns and precautions for working at significant height and with high wind velocities
48. Personal protective equipment (ppe) and clothing that must be worn during The inspection, repair and maintenance activity and from where can it be Obtained

Generic Skills (GS)

User/individual on the job needs to know how to:

1. write official notes
2. write the information communicated by the work in-charge
3. write about the technical problems and other conditions of the site as per standards
4. note critical points and testing repair observations
5. write about the condition of equipment
6. fill in all technical forms and data as per guidelines and format
7. read and correctly interpret written sentences and paragraphs
8. read and correctly interpret units of metric system for all measurements
9. read and correctly interpret the process required for performing work
10. read and correctly interpret the rules and methods from policy and procedure documents
11. read and correctly interpret equipment manuals to identify operational details and specifications
12. communicate clearly with team and other staff members
13. discuss task lists, schedules and activities
14. interact with the junior engineer to clarify instructions and doubts
15. make appropriate work-related judgments based on policy and procedures
16. identify complex problems and review related information to develop and evaluate solutions
17. follow organizations rule-based decision-making process
18. plan and organize tasks to meet quantity and quality specifications and targets
19. build customer relationships and use customer centric approach in taking decisions



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20. identify problems and review related information to develop and evaluate options and implement solutions
21. monitor problem solving to take corrective action with individuals and organizations
22. analyse problems and changes in conditions, operations, and the environment to solve problems
23. analyse the problems in the equipment to identify root causes
24. collect information and technical data and define process for equipment testing
25. analyse the problem and required actions to resolve them to determine which issues are to be escalated, and to whom
26. develop holistic and comprehensive profile of products based on segregated discrete process stages
27. critically evaluate operation parameters in relation to product features intended to ensure a match



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for repair and maintenance of Power Transmission lines</i>	8	25	-	-
PC1. identify various types of circuits and their components correctly	1	3	-	-
PC2. identify and acquire correct tools, equipment and instruments required for various aspects of repair, maintenance, assessment and inspection of transmission lines and components	1	2	-	-
PC3. identify hazards in trimming trees such as limits of approach, public safety and step and touch potential access	1	2	-	-
PC4. conduct site inspection for emergency cases following established procedures	1	2	-	-
PC5. climb tower while following all necessary safety procedures and using ppe	-	2	-	-
PC6. identify various types of circuits accurately	1	3	-	-
PC7. ensure the tools and equipment are well maintained, calibrated and approved for use	-	2	-	-
PC8. use transmission line tools, equipment and hardware in line with job requirements for maintenance operations	1	2	-	-
PC9. prepare and maintain the work area as per procedure or operation specification	1	3	-	-
PC10. obtain work permit to proceed with work from the appropriate personnel in accordance with the organizational procedure	1	2	-	-
PC11. switch off, isolate, discharge and earth (side) line cables	-	2	-	-
<i>Perform maintenance of power transmission lines as job specifications and requirements</i>	17	34	-	-
PC12. perform off-line overhead line maintenance procedure	1	3	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. perform off-line underground line maintenance procedure	1	2	-	-
PC14. ensure pole dismantling and re-setting procedure is carried out as per standard procedure, where required	1	2	-	-
PC15. install components on transmission lines including gang operated air brake switches for transmission lines, controlled breakers, ground switches, capacitor stations, insulator pressure washing, submarine and underground transmission cable, grid interconnections, etc.	2	3	-	-
PC16. select and use test equipment such as tong testers, clip-on meter, multimeters, fault indicators meggers and voltmeters to verify fault and integrity	1	3	-	-
PC17. document switching procedures with all relevant details clearly and accurately	2	2	-	-
PC18. repair conductor by splicing, jointing, using armor rods, line guards, vibration dampers	1	2	-	-
PC19. evaluate the work carried out by the team members and ensure it is as per the standard requirements and provide any feedback in a timely, polite and supportive manner	1	2	-	-
PC20. report any issues along with the remedial actions such as repairs or replacements, and estimated repair time to the relevant authority	1	2	-	-
PC21. replace pole as per standard procedure, where required	1	2	-	-
PC22. carry out guy and anchor replacement on various structure types: wood, steel, various lines voltages	1	2	-	-
PC23. carry out conductor patch and splice repair on single conductor, bundled conductor of various sizes and line voltages	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. replace components such as transformers, ct (current transformer), cvt (capacitive voltage transformer), la (lightening arrestor), breakers, towers, conductors, disconnects, timber or x-arm, conductors, poles, switches, elbows, terminations and insulators safely	1	3	-	-
PC25. replace other line components due to damage or unsuitability as per standard procedure, where required	1	2	-	-
PC26. replace underground cables, as per standard procedures where required	1	2	-	-
<i>Perform post repair and maintenance activities</i>	5	11	-	-
PC27. restore the system to normal operating status by using switching procedures	1	2	-	-
PC28. identify problems that are not within one's own capability or limits of authority and seek support from authorized persons for resolving the same	1	2	-	-
PC29. leave the work area in a safe and tidy condition on completion of the repair and maintenance activities	-	2	-	-
PC30. refer unresolved job-related problems to the appropriate personnel for support	2	2	-	-
PC31. monitor the problem and keep the supervisor informed about progress or any delays in resolving the problem	1	3	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	PSS/N3403
NOS Name	Perform repair & maintenance of power transmission lines and components
Sector	Power
Sub-Sector	Transmission
Occupation	Operation and Maintenance - Transmission
NSQF Level	4
Credits	9
Version	1.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass 70% aggregate for the QP.



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7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
PSS/N1331.Apply basic health and safety practices for power related work	39	61	0	0	100	10
PSS/N1336.Working effectively with others	34	56	0	0	90	10
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	15
PSS/N3403.Perform repair & maintenance of power transmission lines and components	30	70	-	-	100	65
Total	123	217	0	0	340	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.